

Arduino & Embedded Basics

Program real hardware · Equigrow short notes

Sketch structure

```
const int LED = 13;
void setup() {
  pinMode(LED, OUTPUT);
  Serial.begin(9600);
}
void loop() {
  digitalWrite(LED, HIGH); delay(500);
  digitalWrite(LED, LOW); delay(500);
  Serial.println(analogRead(A0)); // 0-1023
}
```

Key concepts

Concept	Meaning
Digital I/O	HIGH/LOW: LEDs, buttons
Analog in (ADC)	0-1023 reading of 0-5 V (on Uno)
PWM	analogWrite 0-255: LED brightness, motor speed
UART / I2C / SPI	ways to talk to sensors and modules
Interrupts	react instantly to a pin change

Sensors & motors

- Ultrasonic (HC-SR04): distance = time × 0.034 / 2 cm.
- Servo: angle with the Servo library. DC motors need a driver (e.g. L298N); never power motors from the board pin.
- Use a pull-up/pull-down resistor on buttons, or INPUT_PULLUP.